

Development of Joint Bluetooth and mioty Localization Module

The LIKE is developing new methods for low-cost IoT localization. One promising approach combines Bluetooth Low Energy (BLE) and mioty within the 2.4 GHz band. This method leverages BLE's low energy consumption alongside mioty's long-range capabilities, all on the same hardware. BLE signals are transmitted at short intervals for precise tracking, while mioty signals are sent less frequently, enabling device localization even when no BLE receiver is nearby.

As part of a research project, suitable hardware and corresponding firmware for a combined Bluetooth and mioty system shall be developed, implemented, and tested. The system shall be designed for ultra-low power consumption, enabling operation with a CR2032 battery.

The tasks are:

- Review of the existing work on joint Bluetooth and mioty hardware
- Development of a new system concept
- Development of the required hardware
- Development of the required firmware
- Assembly of the hardware
- Test of the complete system

All findings will be documented in a written report. Additionally, the source code must be thoroughly commented for clarity. The student must also adhere to the guidelines of the German Research Foundation (DFG) for Good Scientific Practice.

Algorithms for Joint Bluetooth and mioty Localization

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As part of a research project, algorithms for the joint localization of Bluetooth and mioty shall be developed. This e.g. includes the interfacing to the Bluetooth and mioty receivers as well as the required signal processing for the localization.

The tasks are:

- Review of the existing work on joint Bluetooth and mioty localization
- Development of a new system concept and the required interfaces
- Implementation of the required software
- Test of the complete system

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Visualization of a Joint Bluetooth and mioty Localization System

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As part of a research project, a web-based interface will be developed to visualize the positions of the combined Bluetooth and mioty transmitters. The software will be built using the well-established Leaflet framework, with necessary custom extensions.

The tasks are:

- Review of the existing work on the visualization of the position of devices on a map
- Development of a suitable system concept
- Interfacing with the localization backend
- Implementation of the required software
- Test of the complete system

All findings will be documented in a written report. Additionally, the source code must be thoroughly commented for clarity. The student must also adhere to the guidelines of the German Research Foundation (DFG) for Good Scientific Practice.